Liquidity-Aware Smart Execution for Desk Efficiency

Microstructure Signals Across Manual Trading and Peak Algo

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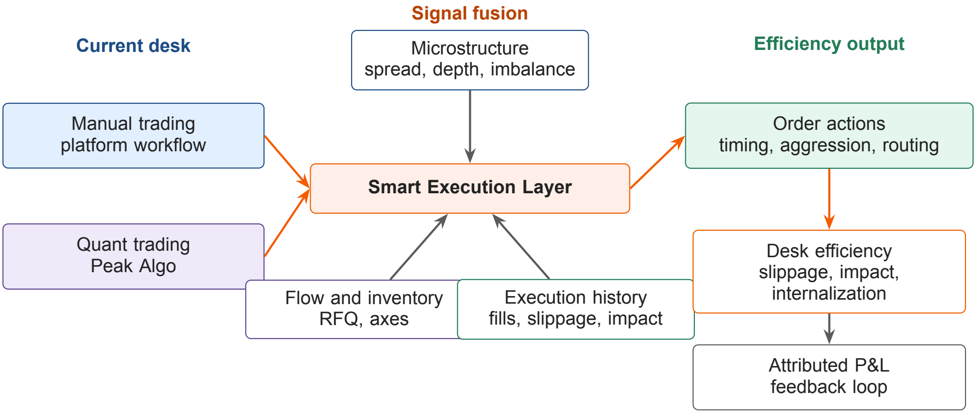
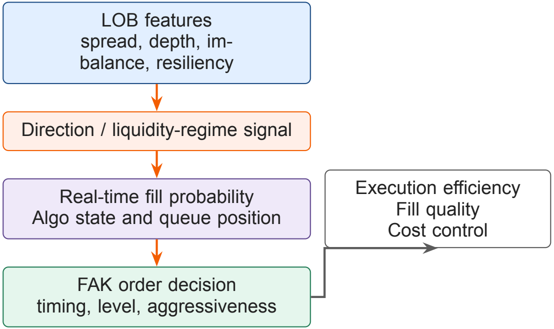
Smart Execution as a Desk Efficiency Layer



FAK Order Execution Efficiency



Case 1: favorable direction

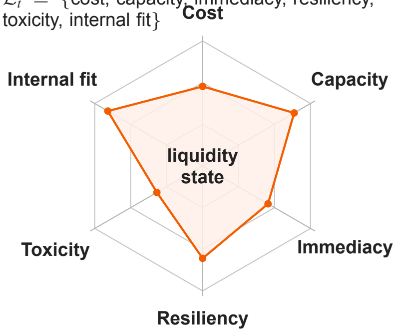
- Direction supports the order side.
- Expected fill probability is higher.
- Execution efficiency: lower slippage and lower market impact.

Case 2: adverse direction

- Direction works against the FAK order.
- Revise fill expectation and aggressiveness.
- Improve success rate and reduce unhedged risk.

Liquidity is the True State Variable

$\mathcal{L}_t = \{$cost, capacity, immediacy, resiliency, toxicity, internal fit$\}$



How to read the state

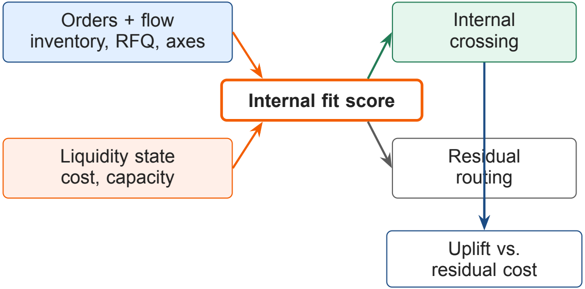
- **High capacity + fast resiliency:** raise FAK confidence.
- **High cost or toxicity:** revise fill expectation and aggression.
- **High internal fit:** cross internally before external routing.
- **Weak state:** delay, slice smaller, or hedge earlier.

Design rule

Price is the observation. Liquidity is the control state.

Source: Lehalle and Laruelle; Cont and de Larrard; Bacry et al.; LOBFrame / LOB benchmark studies.

Microstructure for Internal Flow



Research-backed decision

- Price external liquidity by state, not quote alone.
- Estimate internal fit from inventory, RFQ and axes.
- Route residual flow when depth and resiliency are favorable.
- Prove uplift against residual external cost.

0.01 bp as Attributed Execution P&L

Base-case value

90M

annual target from 0.01 bp saving

- 2025 flow: 45tn.
- Average duration: 2 years.
- $45tn \times 2 \times 0.01bp \approx 90M$.

Attribution stack

- **Internal crossing:** avoid spread and external market impact.
- **Liquidity timing:** trade when depth and resiliency are favorable.
- **Adverse-selection control:** reduce hedge leakage and bad fills.

Proof metrics

- Slippage vs. arrival / decision price.
- Post-trade impact and hedge delay.
- Internalization uplift vs. residual external cost.